Group3 Summary

研究問題:

經過第一次的程式運行後發現資料中為零的數值過多且有些不合理的異常值。而過多異常值有可能影響數據而導致分析結果判斷錯誤。所以我們想要對資料先進行處理，同時加入長條圖以及熱力圖來分析結果。

研究方法:

1. 設定各指標的閥值。
2. 對資料進行分組分別為:Tiny、Low、Normal、High、Excessive、Over
   後續用來觀察刪除了多少資料。
3. 刪除資料裡的異常值。
4. 透過長條圖以及熱力圖來分析各個指標與糖尿病的相關性。

主要圖表:

設定各指標的閥值

norm.csv

```csv
limit,Glucose,BloodPressure,Insulin,BMI
Tiny,70,30,-0.1,0
Low,70,60,16,18.5
Normal,99,80,166,25
High,125,90,166,30
Excessive,200,200,300,60
```

```python
norm_dict = {}

for col in norm.columns[1:]:  # 跳過 limit
    norm_dict[col] = norm[col].tolist()

print(norm_dict)
```

```
{'Glucose': [70, 70, 99, 125, 200], 'BloodPressure': [30, 60, 80, 90, 200]
```

資料分群以及刪除異常值

```python
def classify(value, thresholds):

    tiny, low, normal, high, excessive = thresholds

    if value <= tiny:
        return "Tiny"
    elif value <= low:
        return "Low"
    elif value <= normal:
        return "Normal"
    elif value <= high:
        return "High"
    elif value <= excessive:
        return "Excessive"
    else:
        return "Over"
classification_df = pd.DataFrame()

for col in norm_dict.keys():
    classification_df[col] = df[col].apply(
        lambda x: classify(x, norm_dict[col])
    )

print(classification_df.head())
```

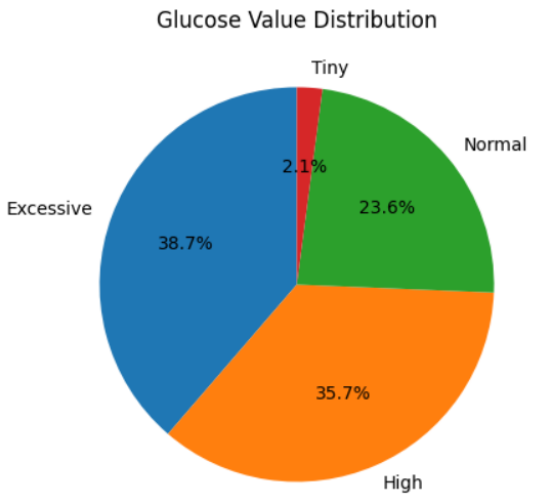
```python
clean_df = df.copy()

mask = pd.Series(True, index=df.index)

for col in classification_df.columns:
    mask &= ~classification_df[col].isin(["Tiny","Over"])

clean_df = df[mask]

print("Before:", df.shape)
print("After:", clean_df.shape)
```

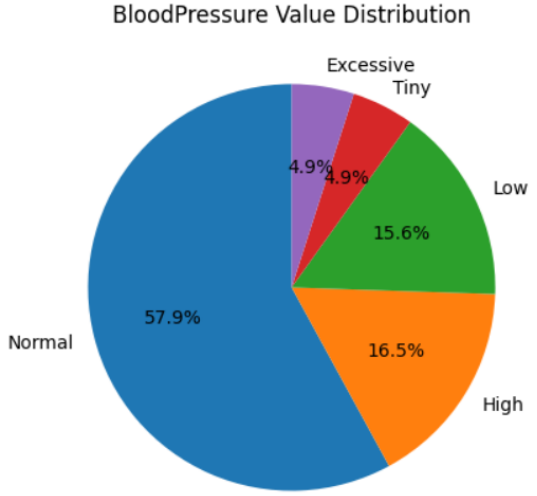
將預處理後的資料重新計算平均、標準差等

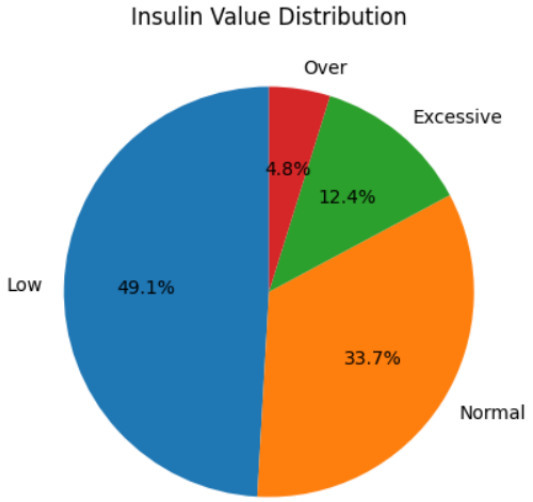
	Mean Before	Mean After	Std Before	Std After	Min Before
Glucose	120.89	121.01	31.97	29.18	0.0
BloodPressure	69.11	72.58	19.36	11.96	0.0
Insulin	79.80	65.76	115.24	79.06	0.0
BMI	31.99	32.27	7.88	6.70	0.0

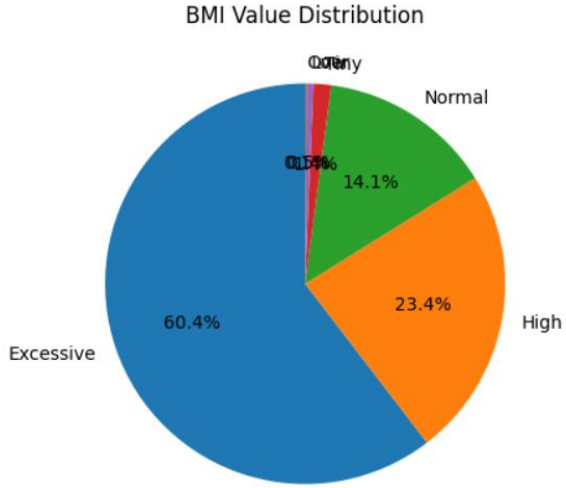
	Min After	Max Before	Max After
Glucose	71.0	199.0	199.0
BloodPressure	38.0	122.0	122.0
Insulin	0.0	846.0	300.0
BMI	18.2	67.1	59.4

圓餅圖(數據分組)

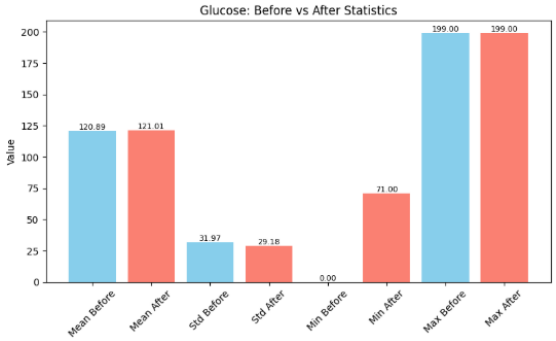

Glucose Value Distribution
Tiny
Normal
Excessive
High
2.1%
23.6%
38.7%
35.7%

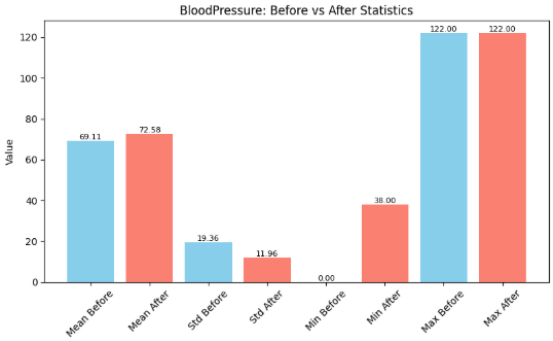

BloodPressure Value Distribution
Excessive
Tiny
Low
Normal
High
4.9%
4.9%
15.6%
57.9%
16.5%

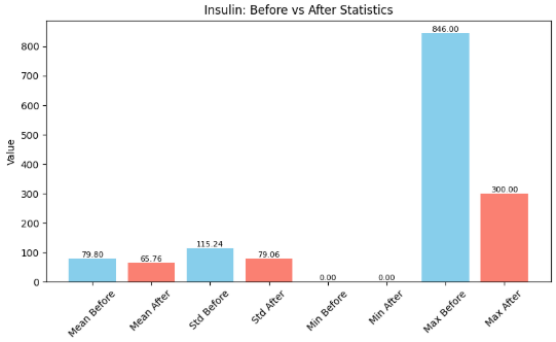

Insulin Value Distribution
Over
Excessive
Low
Normal
4.8%
12.4%
49.1%
33.7%

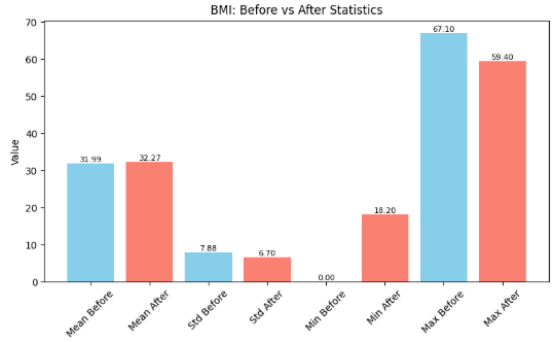

BMI Value Distribution
Normal
High
Excessive
14.1%
23.4%
60.4%

長條圖

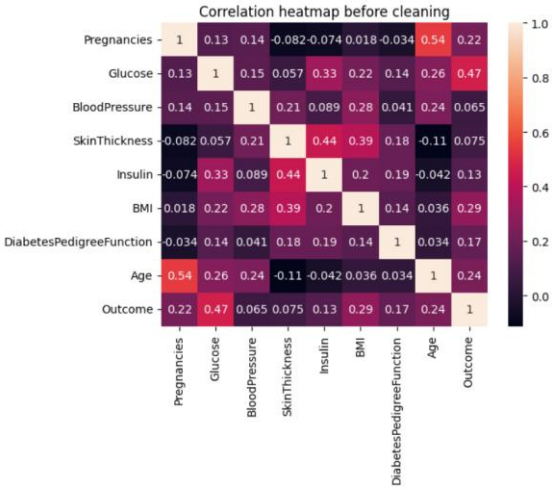

Glucose: Before vs After Statistics
Value
120.89
121.01
31.97
29.18
0.00
71.00
199.00
199.00
Mean Before
Mean After
Std Before
Std After
Min Before
Min After
Max Before
Max After

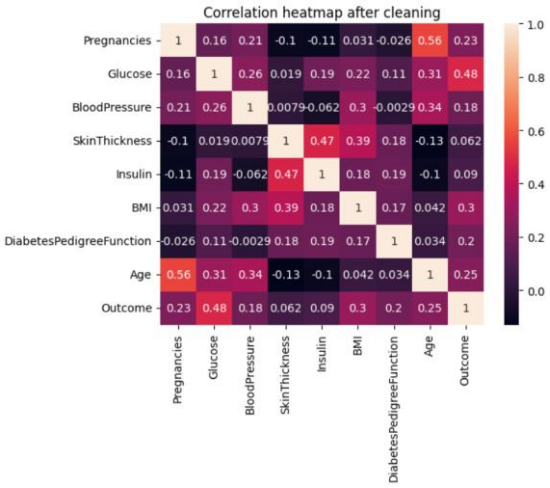

BloodPressure: Before vs After Statistics
Value
69.11
72.58
19.36
11.96
0.00
38.00
122.00
122.00
Mean Before
Mean After
Std Before
Std After
Min Before
Min After
Max Before
Max After


Insulin: Before vs After Statistics
Value
79.80
65.76
115.24
79.06
0.00
0.00
846.00
300.00
Mean Before
Mean After
Std Before
Std After
Min Before
Min After
Max Before
Max After


BMI: Before vs After Statistics
Value
31.99
32.27
7.88
6.70
0.00
18.20
67.10
59.40
Mean Before
Mean After
Std Before
Std After
Min Before
Min After
Max Before
Max After

熱力圖





重要發現:

原始資料中的 0 值可能為缺失值而導致平均值被低估。而醫學上葡萄糖、血壓、胰島素不會為 0，可以在長條圖看到，在經過預處理後始得結果更接近實際發生狀況減少不合理。資料刪減部分可以透過圓餅圖觀察各個指標的資料刪減以及分組狀況。

在熱力圖的部分清理後的數據消除了異常值的干擾，強化了各指標對糖尿病的解釋權重。其中葡萄糖是最為明顯的正相關 0.48 其次為 BMI 以及年齡。而在血壓的部分原本為趨近於無相關 0.065 上升至 0.18，得血壓跟糖尿病的關聯性得以正確建立。

本次分析透過資料預處理後讓我們得到了相較於原始資料更接近真實的數據。使的分析結果能夠更為真實，也透過熱力圖的反應得知預處理有發揮成效，讓結果更有參考價值。